Life cycle assessment of different energy technologies
Environmental Analysis & Ecology

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1 Photovoltaic Park (Matteo)

1.1 Paper 1 [1]

Title: Estimating the environmental footprint of a grid-connected 20 MWp photovoltaic system
Authors: Artúr Szilágyi, Gyula Gróf
Year: 2020

The LCA in this paper is intended to support the planning and eco-design of photovoltaic plants by estimating their environmental impact during the early planning phase. The goal of the paper is twofold: first, to assess the environmental impacts of the power plant using the Product Environmental Footprint (PEF) methodology released by the Official Journal of the European Union [2] and second, to compare the results of the simplified life-cycle model with those of the detailed model.

The study is comparative because it compares the simplified LCA with the detailed LCA, as well as the PV power plant with the Hungarian grid mix.

The function of the system is to generate AC electricity and deliver it to the grid, and the functional unit is 1kWh of AC electricity. The estimated lifetime of the plant is of 25 years, and the expected lifetime production is 550 GWh.

The system boundary is cradle-to-grave, but with two model variants. The detailed model applies a 99% cut-off criterion, meaning that it aims to include more than 99% of the total materials and energy required for the construction of the power plant, and includes PV modules, support structure, inverters, transformers, cables, auxiliary infrastructure such as roads, fence and buildings, transport to site, diesel and electricity for installation, packaging, site preparation, installation waste, operation and maintenance (O&M), replacement of faulty components, and end-of-life transport and waste treatment. The excluded processes are capital equipment and machinery life cycles, commuting, administration, marketing, R&D, and the energy needed for decommissioning, because it could not be reliably estimated. The simplified model takes into account only the dominant contributors of the environmental impact, which are PV panels, mounting structure, inverters, and cables. Installation, transport and most O&M processes are neglected; O&M is reduced to replacement of modules and inverters, and end-of-life is represented mainly by waste cable treatment. This simplified boundary still captures most impacts.

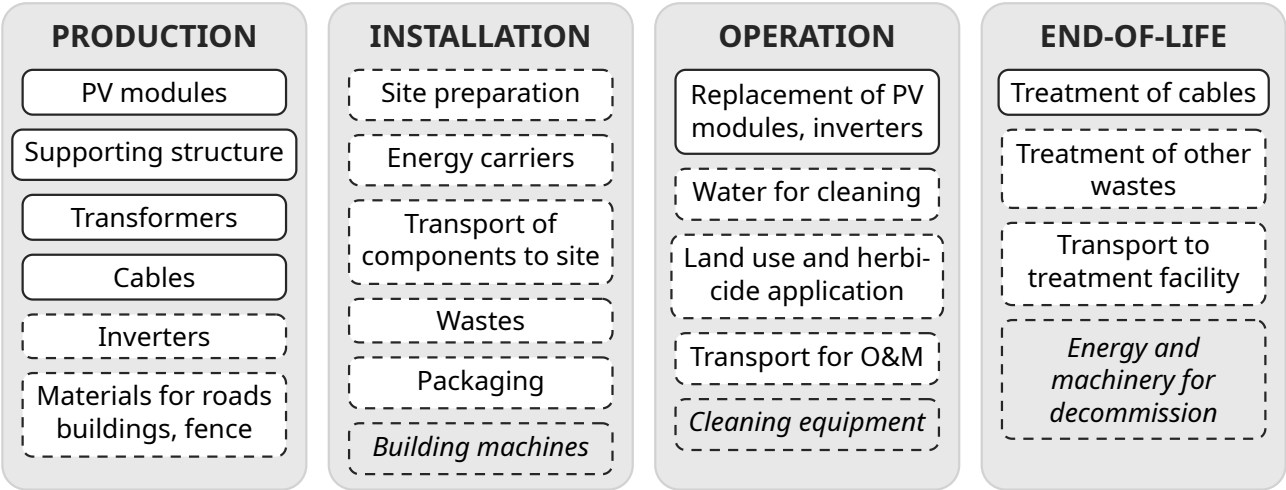

Figure 1: Life-cycle stages considered in the study. Processes with white background are considered in the detailed model, while processes with dashed borders are excluded from the simplified model. Processes with italic font are excluded from both models.

The inventory requires material quantities, component numbers, component masses, transport distances, diesel and electricity use, replacement rates, degradation rate, lifetime electricity generation and end-of-life treatment assumptions. Primary data is retrieved by the commissioner of the power plants, while the secondary data, mainly raw material, equipment, and fuels data, is taken from the latest available ecoinvent database version (v3.5) released in 2018. Most of the data is measured or calculated, while only transported quantity and end-of-life assumptions are estimated. The case study is located in Felsőzsolca, Hungary, and the data is representative of the Hungarian plant and European averages. Regarding the precision and the completeness of the data, the secondary PV-module dataset was checked against a review and found to be in the same range, but transport and end-of-life data remain

uncertain. The study did not model the decarbonisation of the grid over the 25-year lifetime, limiting the comparison with the Hungarian grid mix.

The impact categories are grouped by damage type. Climate change is represented by global warming 100a. Ecosystem quality includes freshwater and terrestrial acidification, freshwater ecotoxicity, freshwater eutrophication, marine eutrophication, and terrestrial eutrophication. Human health includes carcinogenic effects, ionising radiation, non-carcinogenic effects, ozone layer depletion, photochemical ozone creation, and respiratory effects from inorganics. Resources include fossil fuel depletion, land use, minerals and metals depletion, and water depletion.

Table 1: Results of the case study vs. simplified model in PEF pts and impact coverage in (%) by damage category

Damage category	A: Simplified model	B: Detailed model	Share (A/B)
Climate change	8.67×10^3 PEF pt	9.47×10^3 PEF pt	92%
Ecosystem quality	2.67×10^4 PEF pt	2.95×10^4 PEF pt	91%
Human health	1.18×10^5 PEF pt	1.36×10^5 PEF pt	87%
Natural resources	2.23×10^5 PEF pt	2.38×10^5 PEF pt	94%
Total	3.76×10^5 PEF pt	4.13×10^5 PEF pt	91%

The total detailed result is equivalent to 7.5×10^{-4} PEF pt/kWh based on the 550 GWh lifetime output. The converted GWP is 6.57 g CO_2-eq/kWh for the detailed model and 6.02 g CO_2-eq/kWh for the simplified model.

The main contributors are PV production at 47.0% and steel structure at 24.3%, followed by the inverter production at 6.6%, cable production at 5.5%, and other smaller processes. Thus, PV modules and the mounting structure dominate the footprint.

1.2 Paper 2 [3]

Title: Life Cycle Assessment of a ground-mounted 1778 kW_p photovoltaic plant and comparison with traditional energy production systems
Authors: Umberto Desideri, Stefania Proietti, Francesco Zepparelli, Paolo Sdringola, Silvia Bini
Year: 2012

This paper is intended to assess the environmental impact of generating electricity using a large ground-mounted photovoltaic system. The purpose is merely publication-oriented and it is based on a comparative analysis, as electricity generated by the ground-mounted PV system is compared with electricity produced from conventional energy sources, namely the Italian electricity mix, natural-gas turbine electricity, oil-fired electricity, and hard-coal electricity. It also compares different end-of-life scenarios for PV modules and the PV plant with the Swiss PV mix.

The product function is electricity production from a ground-mounted photovoltaic plant. Two functional units are used: 1 kWp of installed power for the assembly inventory and 1 kWh of electricity produced for the full life-cycle comparison. The energy production is estimated over a 25-year lifetime, and the installed peak power of the plant is 1778 kWp.

The system boundary is cradle-to-grave. It includes land preparation, fence installation, electrical substations of precast concrete, support structures, wiring, instrumental apparatus for electrical network connection, PV module production, module transportation to the construction site, module installation, operation, maintenance, decommissioning and disposal/recycling. In particular, the study takes into account the soil preparation, the installation of fence and electrical substations of low and medium voltage, the mounting of support structures, also with reference to hot dip galvanizing process, the production of modules, their installation, the wiring apparatus and the network connection. The transport of all components to the installation site is considered for each stage that is examined. The end of life scenario of the plant is also evaluated.

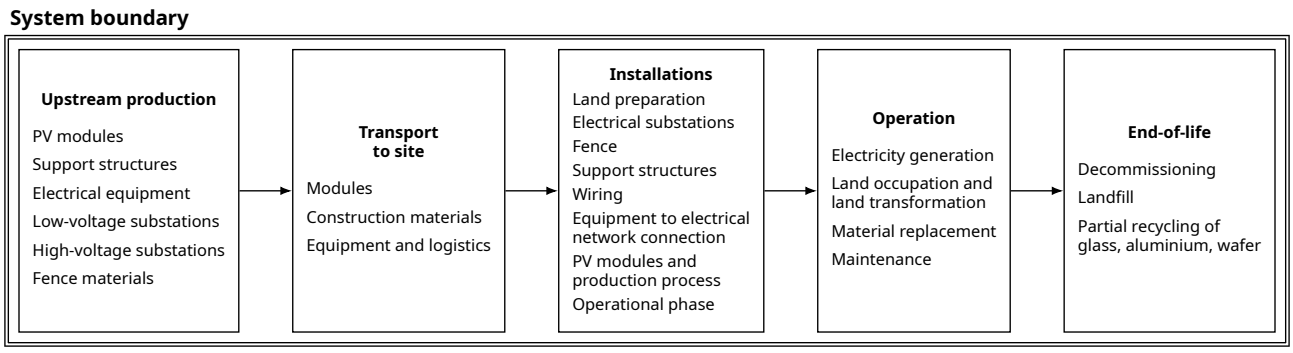

Figure 2: System boundary of the ground-mounted photovoltaic plant

The required inventory data include diesel consumption, transport distances, raw materials, component composition, module manufacturing data, inverter and transformer composition, cable material shares, construction-machine use, maintenance replacements, and end-of-life recycling/disposal assumptions. A large amount of the data is retrieved from the documentation of TerniEnergia, and some other is measured on-site. The secondary data is taken from literature and SimaPro databases. The plant was built in November 2009 and became operational in April 2010, when PV modules were still under study. The power plant is located in Marsciano, Italy, and the study mentions that the PV modules were manufactured in China and transported through France to Italy. The periodic monitoring of the construction site during the building and operating phases provides high data precision, resulting in no missing data and therefore no need to fill data gaps. The main uncertainty is the production data of the modules, as it is stated that the exact information of the modules was not directly provided so the literature was adapted relying on another publication.

The impact assessment method used is Eco-indicator 99, which groups impacts into three main damage categories: Human Health, Ecosystem Quality, and Resources. Human Health includes carcinogens, respiratory organics, respiratory inorganics, climate change, radiation, and ozone layer depletion. Ecosystem Quality includes ecotoxicity, acidification/eutrophication, and land use. Resources include minerals and fossil fuels.

With a total of 177 Pt, the LCIA shows that PV modules are the largest contributor in the assembly phase, scoring 118 Pt, followed by the electrical network connection apparatus with 31.4 Pt and wiring with 18 Pt.

For the full life cycle, the results differ based on the module end-of-life scenario. If the modules are landfilled the score is 7.79 mPt/kWh, if glass and aluminium are partially recovered the score is 7.68 mPt/kWh, and including the wafer recovery it is 7.34 mPt/kWh.

The main climate and energy indicator is GWP100, with a value of 88.743 g CO_2/kWh, or 0.106 kg CO_2-eq/kWh.

1.3 Comparison of Paper 1 and Paper 2

Despite the fact that both papers carry out the LCA of ground-mounted PV plants, each with a 25-year lifespan, and apply the cradle-to-grave approach, their LCA studies are partially comparable. They can be compared in terms of technological aspects as they analyse the same function, that is the production of electricity by PV panels, considering 1 kWh of electricity generated over the life cycle as the functional unit, including modules, structural elements, electrical equipment, transportation, operation and maintenance (O&M), and end-of-life management. However, their systems differ in terms of the applied boundaries, assumptions, and LCIA methods.

Paper 1 evaluates a 20 MWp plant installed in Hungary through the PEF method, applying the detailed and simplified models without machinery life cycle assessment and energy for decommissioning. On the other hand, Paper 2 estimates the environmental footprint of a 1778 kWp Italian plant using Eco-indicator 99, taking into account detailed construction site information, shipment of modules from China, O&M, decommissioning, and various recycling options. Moreover, the damage categories differ, as Paper 1 provides the PEF damage categories and Paper 2 presents Eco-indicator 99 damage categories and single score results. With appropriate calculations, the two LCAs can be compared and mainly for common indicators such as the GWP100.

2 Nuclear Power Plant (Yannik)

2.1 Paper 1

Title:
Authors:
Year:

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2.2 Paper 2

Title:
Authors:
Year:

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2.3 Comparison of Paper 1 and Paper 2

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3 Coal-fired Power Plant (Anastasia)

3.1 Paper 1 [4]

Title: Life cycle and environmental impact assessment of coal-fired power plants in Bangladesh
Authors: Khalid Shaifullah Mahmud, Md. Rasel Ahmed, Sajal Ahmed, Jahidul Islam Jihan, Mst. Tajnin Nahar Tonni, Md. Golam Kibria, Md. Rabiul Islam Sarker
Year: 2025

3.2 Paper 2 [5]

Title: Environmental impact assessment of a remote coal-fired power plant in Central Kalimantan: A life cycle assessment approach and mitigation strategies
Authors: Yohanes Christda Batista, Subhan Hasisi, Muchamad Arief Dharmawan, Bagus Adi Putra, Sunu Herwi Pranolo, Muflih Arisa Adnan
Year: 2025

3.3 Comparison of Paper 1 and Paper 2

3.3.1 Goal and Scope

Question A

Plant 1: Leading Bangladesh towards policy for sustainable energy in growing demand.

Plant 2: Helping reach frameworks such as the Paris Agreement.

Question B

Plant 1: Evaluation of resource use, emissions and impact.

Plant 2: Quantifying existing emissions and identifying strategies for minimizing future emissions.

Question C

Plant 1: Policy makers, energy engineers and fellow researchers.

Plant 2: Fellow researchers and engineers.

Question D

Plant 1: Yes (3 plants).

Plant 2: No.

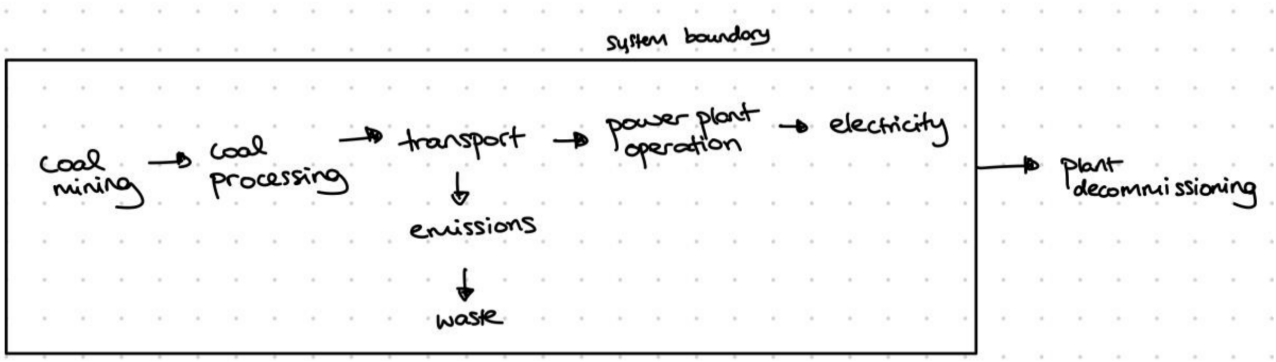
Question E

Plant 1: Electricity generated; 1 MWh.

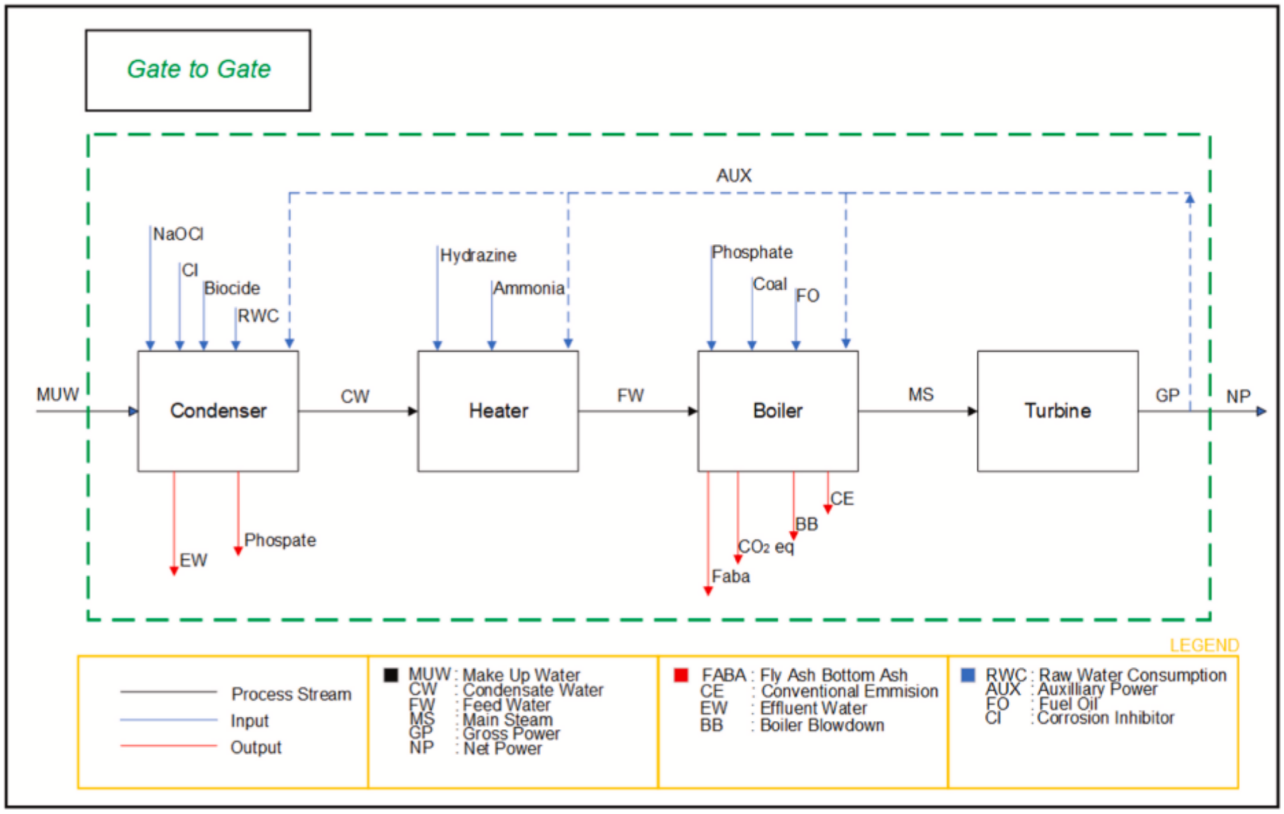
Plant 2: Electricity generated from the power plant; 1 kWh.

Question F

Plant 1:



Plant 2 (Reference paper 2):



Question G

Plant 1: This LCA has three geographical references, all in Bangladesh: Barapukuria (subcritical), Rampal (supercritical), and Payra (ultra-supercritical). It claims high precision and uses modern and state-of-the-art technologies and software that includes OpenLCA, ReCiPe 2016 Midpoint (H), CML-IA baseline, and ILCD 2011 Midpoint. The missing local plant data is substituted using the datasheets: Ecoinvent v3.8 database and ef_secondarydata_202202 for energy supply, material, transport and emission process. The time reference used is 2020–2025 for plant data and 2015–2021 for datasheets. The report states that it follows ISO 14040/14044 transparency and reproducibility standards. This LCA is cradle-to-gate and does not take into account end-of-life, and as such no data for it is presented.

Plant 2: This LCA looks solely into one power plant located in Central Kalimantan, Indonesia. It is precise for the plant itself, as it uses 99% of data from the power plant, in the years 2022–2024, but as such it is hard to use it as an estimate for another place. The data types needed are material inputs (water, coal, chemicals), their amounts, and operational parameters. Where data is missing, Ecoinvent databases are used, and OpenLCA is used as a tool to process the LCA. The assessment is gate-to-gate, and as such is missing upstream processes and infrastructure cycles, for which the gaps are not filled. This LCA can be reproduced in method for other plants, but for the data used only for this plant, as it relies heavily on existing datasheets.

Question H

Plant 1: Global Warming Potential (GWP), Acidification Potential (AP), Eutrophication Potential (EP), Particulate Matter Formation, Human Toxicity, Freshwater Ecotoxicity, Water consumption / Resource depletion.

Plant 2: Global Warming Potential (GWP), Acidification Potential (AP), Eutrophication Potential (EP) and Ozone Depletion.

3.3.2 Results of Life Cycle Impact Assessment

Question A

Plant 1: GWP (Barapukuria): 765–767 kg CO_2-eq/MWh, GWP (Rampal): 826–828 kg CO_2-eq/MWh, GWP (Payra): 841–843 kg CO_2-eq/MWh, AP: 1.5–3.2 kg SO_2-eq/MWh, EP: 0.15–0.45 kg PO_4-eq/MWh, PM Matter Formation: 0.8–1.5 kg $PM_{2.5}$-eq/MWh, Water consumption: 1.2–2.5 m^3/MWh.

Plant 2: GWP: 1.29 kg CO_2-eq/kWh, AP: 2.21×10^{-3} kg SO_2-eq/kWh, EP: 1.31×10^{-2} kg PO_4-eq/kWh, Ozone Depletion: 3.04×10^{-8} kg CFC-11-eq/kWh.

Question B

Plant 1: The goal and scope was achieved, as the three planned power plants were assessed and evaluated with the same methods. The limitations of this LCA are that it excludes plant construction and its end-of-life (decommission-

ing), it relies heavily on secondary datasheets, and it is assumed that coal consumption across the three locations are the same.

Plant 2: The goal and scope was achieved, as the results were measured and quantified in kWh. The limitations of this report are the missing parts of infrastructure, which limits full sustainability assessment. It is a good study for the particular power plant examined, but is not viable when looking at the industry in general.

Between both of the reports, the results are not comparable. The units are different, MWh and kWh, but the conversion between the units is easy, and as such it does not cause a problem when it comes to comparison. What does, is that the first LCA considers cradle-to-gate and the second considered gate-to-gate.

A LCA Questionnaire

A.1 Goal and scope

A.1.1 Goal of LCA

a) What is the intended application of the LCA study?

b) What is the purpose of the LCA study?

c) Who is the intended audience of the LCA report?

d) Is the LCA study based on a comparative analysis?

A.1.2 Scope of LCA

e) What is the function(s) of the product and which functional unit has been applied?

f) Plan and describe the system boundaries of the LCAs using a process flow diagram showing the processes, their relationships, and the function of the system.

g) What data and type of data are needed to calculate inventory results and selected impact assessment categories?

- Data quality requirements, assumptions and limitations
- Data acquisition: Is the data measured, calculated or estimated? How much of the data required is primary data (in %) and how much data is taken from literature and databases (secondary data)?
- Time-reference: When was the data obtained and have there been any major changes since the data collection that might affect the results?
- Geographical reference: For what country or region is this data relevant?
- Is the secondary data from literature or databases representative for state-of-the-art-technology or for older technology?
- Precision: Is the data a precise representation of the system?
- Completeness: Are any data missing? How are data gaps filled?
- Representativeness, consistency, reproducibility: Is the data representative, consistent and can it be reproduced?

h) What impact categories are used?

A.2 Results of Life Cycle Impact Assessment (LCIA)

a) State the results of the different impact categories in numbers.

b) Interpret the two case studies by considering the following:
Results are checked and evaluated to check if they are consistent with the goal and scope definition. This includes: identification of significant issues (e.g. inventory elements or impact categories missing) and evaluation/interpretation of results (e.g. comparability)

B References

References

[1] A. Szilágyi and G. Gróf, "Estimating the environmental footprint of a grid-connected 20 MWp photovoltaic system," *Solar Energy*, **197**, pp. 491–497, 2020, ISSN: 0038-092X, DOI: 10.1016/j.solener.2020.01.028, https://www.sciencedirect.com/science/article/pii/S0038092X20300360. (accessed 2026-04-25).

[2] European Commission, "Commission Recommendation of 9 April 2013 on the use of common methods to measure and communicate the life cycle environmental performance of products and organisations," *Official Journal of the European Union*, L 124, **56**, no. 2013/179/EU, pp. 1–210, Apr. 9, 2013, https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2013:124:FULL:EN:PDF. (accessed 2026-04-25).

[3] U. Desideri, S. Proietti, F. Zepparelli, P. Sdringola, and S. Bini, "Life cycle assessment of a ground-mounted 1778 kWp photovoltaic plant and comparison with traditional energy production systems," *Applied Energy*, **97**, pp. 930–943, 2012, Energy Solutions for a Sustainable World - Proceedings of the Third International Conference on Applied Energy, May 16-18, 2011 - Perugia, Italy, ISSN: 0306-2619, DOI: 10.1016/j.apenergy.2012.01.055, https://www.sciencedirect.com/science/article/pii/S030626191200061X. (accessed 2026-04-25).

[4] K. S. Mahmud, M. R. Ahmed, S. Ahmed, et al., "Life cycle and environmental impact assessment of coal-fired power plants in bangladesh," *Energy Conversion and Management: X*, **28**, p. 101 325, 2025, ISSN: 2590-1745, DOI: https://doi.org/10.1016/j.ecmx.2025.101325, https://www.sciencedirect.com/science/article/pii/S259017452500457X. (accessed 2026-05-02).

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